

Yellow Markets Resolution Logic

Comprehensive Technical Analysis, File Changes Manifest & Project Estimation

Document Purpose: Technical breakdown of the 5-page "Yellow Markets Resolution Logic" specification covering Shopper Storefront, Merchant Panel, Admin Panel (@Help / @Acct), Dispute Isolation in Payouts, and the Multi-Party Email Subsystem.

1. Executive Summary & Core Objectives

The objective of this project is to implement an end-to-end, multi-tier Order Resolution & Dispute Management System across Yellow Markets. It introduces structured decision-making for Merchants (Refunds & Replacements), an escalation dispute avenue for Shoppers, strict role boundaries for Admin Support (@Help) and Finance (@Acct), automated payout withholding during active disputes, and a coordinated 16-event email trigger subsystem.

Key Actors & Responsibilities

- **Shopper:** Initiates tickets directly from "My Orders" selecting order, product, category, and priority. If dissatisfied with a merchant denial, raises a formal **Resolution Request**.
- **Merchant:** Evaluates shopper issues. Can approve/deny refunds or approve/deny replacements. For approved replacements, selects fulfillment route (Own Delivery, Self Pickup, or YM Delivery) and marks replacement completion.
- **@Help (Customer Support Admin):** Coordinates ticket progress, assigns refund requests to @Acct, mediates disputes, approves/denies escalations, and issues closures (standard Close and Close Final).
- **@Acct (Finance / Accounts Admin):** Processes monetary refunds directly deducted from the Merchant's 15-day hold back sales reserve, and sets status to **Done**.

2. Ticket Status Lifecycle & Decision Buttons

The system defines six definitive lifecycle states and specialized actor-gated decision controls:

Status	Badge / Code	Lifecycle Meaning & Actor Permission
Open	STATUS 0: OPEN	Initial state for any new ticket submitted by Shopper or Merchant.
Processing	STATUS 1: PROCESSING	Active processing by @Help or @Acct (e.g. accounting refund execution, dispute evaluation, YM delivery add-on).
Done	STATUS 2: DONE	Restricted to @Acct only once refund/payment deduction from 15-day hold back sales is complete.
Close	STATUS 3: CLOSE	Ticket closed by @Help . Shopper retains the right to trigger a Resolution Request .
ReOpen	STATUS 4: REOPEN	Triggered exclusively by Shoppers clicking "Resolution Request" to dispute a Merchant denial.
Close (Final)	STATUS 5: CLOSE (FINAL)	Final closure by @Help following dispute evaluation. No further reopen options exist.

Decision Controls & Dropdown Specifications

- **Merchant Refund Buttons:** Refund Approved, Refund Denied.
- **Merchant Replacement Buttons:** Replacement Approved, Replacement Denied, and Replacement Completed (conditionally displayed only after approval and delivery option selection).
- **Replacement Delivery Options Dropdown:** Own Delivery, Self Pickup, YM Delivery.
- **Shopper Dispute Button:** Resolution Request (only enabled on tickets closed after denial).
- **@Help Dispute Decisions:** Resolution Approved (shopper favour ? sends to @Acct), Resolution Denied (merchant favour ? Close Final).

3. Core Workflows (The 16 Specification Scenarios)

Phase A: Ticket Creation & Messaging

- **Shopper Ticket Raising:** From "My Orders", Shopper clicks "Ticket" and completes: Category (Delivery, Refund, Replacement, Others), Priority (Low, Medium, High, Urgent), Order No., Product Selection, Message, and optional image upload.
- **Email Triggers on Creation:** Merchant receives notification email (#2); @Help receives notification email (#3).
- **Thread Replies:** Merchant replies with text + image upload; Shopper receives email (#5). When Shopper replies, Merchant receives email (#6). When Merchant clicks decision buttons, @Help receives email (#8).

Phase B: Conclusion Workflows

Workflow	Action Sequence	Emails Triggered
Merchant Approves Refund	1. Merchant clicks "Refund Approved". 2. @Help assigns ticket to @Acct (Status ? Processing). 3. @Acct executes refund from Merchant 15-day hold back sales. 4. @Acct marks status ? Done. 5. @Help marks status ? Close.	• Shopper: "We Approve Refund" (#11b) • @Help: "Order Resolution Update" (#11c) • Merchant: "Refund initiated" (#11e) • @Acct: "Refund Request" (#11f) • Merchant: "Closed" (#11j) • Shopper: "You Are Being Refunded" (#11k)
Merchant Replaces (Own / Pickup)	1. Merchant clicks "Replacement Approved" & selects Own Delivery / Self Pickup. 2. Shopper notified replacement is underway. 3. Merchant replaces item & clicks "Replacement Completed". 4. @Help marks ticket status ? Close.	• Shopper: "Product Replacement Underway" (#12c) • Merchant: "Closed" (#12g) • Shopper: "Your Product Replacement Completed" (#12h/13j)
Merchant Replaces (YM Delivery)	1. Merchant selects "YM Delivery Service". 2. @Help assigns ticket to @Acct (Status ? Processing). 3. @Acct prompts Merchant to purchase AddOn delivery service. 4. Merchant purchases AddOn, product is delivered, Merchant clicks "Replacement Completed". 5. @Help marks ticket ? Close.	• @Help: "Order Resolution Update" (#13c) • Merchant: "Closed" (#13i) • Shopper: "Your Product Replacement Completed" (#13j)
Merchant Denies Refund or Replacement	1. Merchant clicks "Refund Denied" or "Replacement Denied". 2. @Help notified and marks status ? Close. 3. Shopper informed. Shopper now has option to file Dispute.	• Shopper: "No Replacement / Refund for Product" (#14c, #15d) • @Help: "No Replacement / Refund for Product" (#14d, #15e) • Merchant: "Closed" (#14f, #15g)
Shopper Escalation: Resolution Request (Dispute)	1. Shopper clicks "Resolution Request" on Closed ticket. 2. Status ? ReOpen. Both @Help & Merchant notified. 3. @Help mediates and decides: • Scenario 1 (Shopper Favour): @Help clicks "Resolution Approved" ? assigns to @Acct ? refund deducted from 15-day hold back sales ? @Acct marks Done ? @Help marks Close (Final). • Scenario 2 (Merchant Favour): @Help marks Close (Final).	• @Help: "Dispute Refund Decision" (#16c) • Merchant: "Shopper Dispute Decision" (#16d) • @Acct: "Refund Request" (#16g) • Merchant & Shopper: "Closed (Final)" (#16k, #16n) • Shopper: "You Are Being Refunded" or "Refund Disapproved" (#16l, #16o)

4. Financial & Payout Isolation Logic (Rules 9 & 10)

Strict Accounting Constraint (Points 9 & 10 of Specification):

1. "An order in dispute will not be displayed within the Manage Transaction Section."
2. "Once the dispute is closed any order/payout will be displayed within the Manage Transaction Section."

In the current codebase, merchant payouts are generated via `getPayoutOrders()` and validated by `checkPayoutEligibility()` in `B2BOrdersModel.php` and `B2BOrdersController.php`.

- **Query Withholding Filter:** Any sub-order or webshop order linked to a resolution ticket in `help_desk` with an active/unresolved status (`status NOT IN (3, 5)`) must be filtered out and hidden from the Merchant Manage Transactions / Payouts table.
- **Automatic Unblocking:** The moment @Help marks the ticket as `Close (3)` or `Close (Final) (5)`, the order automatically re-qualifies for display and payout execution in accordance with the standard 15-day eligibility timeline.

5. Database Schema Migration Plan

The existing `help_desk` table will be modified and enhanced via the following database migration:

Field Name	Data Type	Description / Enum Values
<code>status</code>	<code>TINYINT(1)</code>	0 = Open, 1 = Processing, 2 = Done, 3 = Close, 4 = ReOpen, 5 = Close (Final)
<code>decision_type</code>	<code>VARCHAR(30)</code>	none, refund, replacement, resolution
<code>decision_status</code>	<code>VARCHAR(30)</code>	none, approved, denied, completed
<code>delivery_option</code>	<code>VARCHAR(30)</code>	none, own_delivery, self_pickup, ym_delivery
<code>assigned_role</code>	<code>VARCHAR(20)</code>	merchant, help, acct
<code>dispute_reason</code>	<code>TEXT NULL</code>	Reason provided by Shopper when clicking "Resolution Request".
<code>resolution_notes</code>	<code>TEXT NULL</code>	Internal findings / notes logged by @Help or @Acct.
<code>attachment</code>	<code>VARCHAR(255) NULL</code>	Preserved on initial ticket and enabled on conversation replies.
INDEX	<code>idx_hd_order_status</code>	Compound index on (<code>order_id</code> , <code>status</code>) for high-performance payout filtering.

6. Complete File Change Manifest

File Path	Portal	Specific Changes Required
database/migration_resolution.sql	Database	Create migration script altering help_desk columns, defaults, and indexes.
application/views/components/order/customer_general_orders.php	Shopper	Add "Ticket" action button on order items; embed Ticket Modal with order/product selection.
application/views/components/order/customer_guest_orders.php	Shopper	Add "Ticket" action button on guest order items.
application/controllers/MyProfileController.php	Shopper	Add mandatory field validations on ticket submit, file upload, and resolutionRequest() endpoint.
application/views/myprofile/help_desk_conversation.php	Shopper	Display 6-status badges, render Resolution Request button on closed tickets, allow reply attachments.
merchant/application/views/help_desk_conversation.php	Merchant	Add Refund Approved/Denied buttons, Replacement Approved/Denied buttons, Delivery dropdown modal, Replacement Completed button.
merchant/application/views/help_desk_list.php	Merchant	Update status badges, ticket filtering tabs, and action links.
merchant/application/controllers/UserController.php	Merchant	Add decision endpoints: refund_decision(), replacement_decision(), replacement_completed().
merchant/application/models/B2BOrdersModel.php	Merchant	Update getPayoutOrders() to exclude orders with open dispute resolution tickets.
admin/application/views/help_desk_conversation.php	Admin	Add controls for @Help (Assign to @Acct, Resolution Approved/Denied, Close, Close Final) and @Acct (Mark as Done).
admin/application/views/shopper_help_desk_list.php	Admin	Display new status badges and filter tickets by role (@Help vs @Acct).

File Path	Portal	Specific Changes Required
admin/application/controllers/CustomerController.php	Admin	Add handler methods for assignment to @Acct, dispute decisions, accounting refund completion, and final closure.
admin/application/models/B2BOrdersModel.php admin/application/controllers/B2BOrdersController.php	Admin	Integrate dispute hold checks inside checkPayoutEligibility() and admin payout tables.
application/helpers/resolution_email_helper.php	System-wide	Create centralized email service method dispatching all 16 scenarios with branded HTML templates.
application/language/english/content_lang.php merchant/application/language/english/content_lang.php	Languages	Add English and French UI translations for all buttons, prompts, statuses, and email templates.

7. Implementation Time & Effort Estimation Matrix

#	Module / Feature	Work Description	Hours
1	Database Schema & Migration	Add columns to help_desk, create status mapping, write & test SQL migration script.	3 – 4 hrs
2	Shopper Ticket Raising (My Orders)	Ticket button on order cards, popup modal with compulsory validations (category, priority, product, message), attachment upload.	4 – 5 hrs
3	Shopper Conversation & Dispute	Live status badges, reply with attachment, "Resolution Request" button on closed tickets, dispute submission flow.	4 – 5 hrs
4	Merchant Decision Workflow	Refund Approved/Denied buttons, Replacement Approved modal with delivery options, Replacement Completed button, controller logic.	8 – 10 hrs
5	Admin @Help & @Acct Workflows	Assign to @Acct, 15-day hold back sales refund deduction, Mark Done, Resolution Approved/Denied, Close & Close (Final).	8 – 10 hrs
6	Payout / Transaction Withholding	Modify getPayoutOrders() & checkPayoutEligibility() in Admin and Merchant to hide disputed orders until closed.	4 – 5 hrs
7	Multi-Party Email Engine (16 Triggers)	Centralized email dispatcher helper, Mailjet/SMTP integration, 16 distinct email subjects/templates across Shopper, Merchant, @Help, @Acct.	8 – 10 hrs
8	Localization (EN & FR)	Add translations for all new buttons, modals, badges, and notification strings in English & French language files.	2 – 3 hrs
9	End-to-End Testing & QA	Full cycle simulation of refunds, replacements (all 3 delivery routes), dispute escalation to @Help, accounting mark done, and payout unblocking.	6 – 8 hrs
TOTAL ESTIMATED TIME:			41 – 54 hrs (~5 to 7 days)

8. Phased Rollout Recommendation

- Phase 1 (Foundation):** Database schema updates and Centralized Email Helper setup.
- Phase 2 (Shopper & Merchant Flow):** "My Orders" ticket creation, conversation thread attachments, and Merchant Decision buttons (Refund / Replacement / Delivery dropdown).
- Phase 3 (Dispute & Admin Resolution):** Shopper "Resolution Request" button, @Help dispute mediation, and @Acct refund deduction from 15-day hold back sales.

4. **Phase 4 (Payout Suppression & QA):** Manage Transactions dispute filter, email verification across all 16 scenarios, and language localization.